

Question:1

A movie theater charges different ticket prices. Kids under 5 are free. Seniors (65+) get a discount. Adults pay the standard price. Write a program that takes a person's age and prints their ticket price ("Free", "Discount", or "Standard").

```
#include<stdio.h>
int main()
{
    int age;
    printf("Enter Your Age:\t");
    scanf("%d",&age);
    if(age<=5)
        printf("Free");
    else if(age>65)
        printf("Discount");
    else
        printf("Standard");
    return 0;
}
```

```
Enter Your Age: 4
Free
-----
Process exited after 2.344 seconds with return value 0
Press any key to continue . . .
```

```
Enter Your Age: 70
Discount
-----
Process exited after 5.789 seconds with return value 0
Press any key to continue . . .
```

```
Enter Your Age: 26
Standard
-----
Process exited after 15.47 seconds with return value 0
Press any key to continue . . .
```

Question:2

A traffic light system works in two stages. First, it checks if the power is on (powerOn is 1). If the power is on, it then checks the lightColor ('R', 'Y', 'G') to decide the action: "Stop" for red,

Student Name: _____Wajiha Nazir_____Roll No: _____25K-3021_____

Section: BSE-1B

"Caution" for yellow, "Go" for green. If the power is off, it should always print "Signal Off".

Write the code for this system

```
#include<stdio.h>
int main()
{
    int power, color;
    printf("\n1=Power On\n0=Power Off\nEnter Status:\t");
    scanf("%d",&power);
    if(power==1)
    {
        printf("1.Red\n2.Yellow\n3.Green\nEnter Light Color:\t");
        scanf("%d",&color);
        if(color==1)
            printf("Stop");
        else if(color==2)
            printf("Caution");
        else if(color==3)
            printf("Go");
        else
            printf("Invalid Input");
    }
    else if(power==0)
        printf("Signal Off");
    else
        printf("Invlaid Input");
    return 0;
}
```

```
1=Power On
0=Power Off
Enter Status:  1
1.Red
2.Yellow
3.Green
Enter Light Color:  2
Caution
-----
Process exited after 8.329 seconds with return value 0
Press any key to continue . . .
```

```
1=Power On
0=Power Off
Enter Status:  0
Signal Off
-----
Process exited after 3.034 seconds with return value 0
Press any key to continue . . .
```

Question: 3

Write a program to solve a quadratic equation $ax^2 + bx + c = 0$. Input a, b, c. - Use to calculate the discriminant.

Student Name: _____Wajiha Nazir_____Roll No: _____25K-3021_____

Section: BSE-1B

- If discriminant $> 0 \rightarrow$ two real roots.
- If discriminant $= 0 \rightarrow$ one real root.
- If discriminant $< 0 \rightarrow$ imaginary roots

```
#include<stdio.h>
#include<math.h>

int main()
{
    float a,b,c,D,x1,x2;
    printf("Enter the value of a:\t");
    scanf("%f",&a);
    printf("Enter the value of b:\t");
    scanf("%f",&b);
    printf("Enter the value of c:\t");
    scanf("%f",&c);
    D=((b*b)-(4*a*c));
    if (D<0)
        printf("\nThe roots are imaginary");
    else if(D==0)
    {
        x1=-b/(2*a);
        printf("\n\n\tThe values of x1=\t%f",x1);
        printf("\n\n\tThe roots are real and equal");
    }
    else
    {
        x1=(-b+sqrt(D))/(2*a);
        x2=(-b-sqrt(D))/(2*a);
        printf("\n\n\tThe roots of x1=\t%f",x1);
        printf("\n\n\tThe roots of x2=\t%f",x2);
        printf("\n\n\tThe roots are real");
    }
    return 0;
}
```

```
Enter the value of a:  1
Enter the value of b:  1
Enter the value of c:  1

The roots are imaginary
-----
Process exited after 24.35 seconds with return value 0
Press any key to continue . . .
```

Student Name: Wajiha Nazir Roll No: 25K-3021

Section: BSE-1B

```
Enter the value of a:  1
Enter the value of b: -7
Enter the value of c: 12

      The roots of x1=      4.000000

      The roots of x2=      3.000000

      The roots are real
-----
Process exited after 5.368 seconds with return value 0
Press any key to continue . . .
```

```
Enter the value of a:  1
Enter the value of b: -6
Enter the value of c:  9

      The values of x1=      3.000000

      The roots are real and equal
-----
Process exited after 4.072 seconds with return value 0
Press any key to continue . . .
```

Question:4

An online account login has two security checks. First, it checks if the username is correct. If it is, it then checks the password. If both are correct, it logs the user in. However, if the username is correct but the password is wrong, it should give a specific error: "Incorrect password". If the username itself is wrong, the error should be "Username not found". Write the complete nested logic for this.

Student Name: _____Wajiha Nazir_____Roll No: _____25K-3021_____

Section: BSE-1B

```
#include <stdio.h>
#include <string.h>

int main() {
    char correct_username[] = "admin";
    char correct_password[] = "12345";
    char username[50];
    char password[50];
    printf("Enter username: ");
    scanf("%s", username);
    if (strcmp(username, correct_username) == 0) {
        printf("Enter password: ");
        scanf("%s", password);
        if (strcmp(password, correct_password) == 0) {
            printf("Login successful\n");
        } else {
            printf("Incorrect password\n");
        }
    } else {
        printf("Username not found\n");
    }
    return 0;
}
```

```
Enter username: admin
Enter password: 12345
Login successful
```

```
-----
Process exited after 10.74 seconds with return value 0
Press any key to continue . . .
```

```
Enter username: admin
Enter password: 20453
Incorrect password
```

```
-----
Process exited after 7.623 seconds with return value 0
Press any key to continue . . .
```

```
Enter username: abcdf
Username not found
```

```
-----
Process exited after 4.847 seconds with return value 0
Press any key to continue . . .
```

Question:5

A university has rules for course enrollment. A student can only enroll if they have completed the prerequisite (hasPrerequisite is 1). If they have the prerequisite, then they can enroll if the course is not full (isCourseFull is 0). Write the logic that prints one of these four precise messages: "Enrolled successfully", "Cannot enroll: prerequisite missing", "Cannot enroll: course is full", or "Cannot enroll: prerequisite missing and course is full."

```
#include <stdio.h>

int main() {
    int hasPrerequisite;
    int isCourseFull;
    printf("Enter 1 if prerequisite completed, 0 if not: ");
    scanf("%d", &hasPrerequisite);
    printf("Enter 1 if course is full, 0 if not: ");
    scanf("%d", &isCourseFull);
    if (hasPrerequisite == 1 && isCourseFull == 0)
        printf("Enrolled successfully\n");
    else if (hasPrerequisite == 0 && isCourseFull == 0) {
        printf("Cannot enroll: prerequisite missing\n");
    }
    else if (hasPrerequisite == 1 && isCourseFull == 1) {
        printf("Cannot enroll: course is full\n");
    }
    else if (hasPrerequisite == 0 && isCourseFull == 1) {
        printf("Cannot enroll: prerequisite missing and course is full\n");
    }
    return 0;
}
```

```
Enter 1 if prerequisite completed, 0 if not: 1
Enter 1 if course is full, 0 if not: 0
Enrolled successfully

-----
Process exited after 2.364 seconds with return value 0
Press any key to continue . . .
```

```
Enter 1 if prerequisite completed, 0 if not: 0
Enter 1 if course is full, 0 if not: 0
Cannot enroll: prerequisite missing

-----
Process exited after 7.895 seconds with return value 0
Press any key to continue . . .
```

Student Name: Wajiha Nazir Roll No: 25K-3021

```
Enter 1 if prerequisite completed, 0 if not: 1
Enter 1 if course is full, 0 if not: 1
Cannot enroll: course is full

-----
Process exited after 2.476 seconds with return value 0
Press any key to continue . . .
```

```
Enter 1 if prerequisite completed, 0 if not: 0
Enter 1 if course is full, 0 if not: 1
Cannot enroll: prerequisite missing and course is full

-----
Process exited after 4.368 seconds with return value 0
Press any key to continue . . .
```

Question:6

A smart home system controls appliances based on the room and the desired action. The system first asks for the room ('L' for Living Room, 'K' for Kitchen). Then it asks for the action ('L' for Lights, 'T' for Thermostat).

The specific command depends on the combination:

- Living Room Lights: "Adjusting ambient lighting."
- Living Room Thermostat: "Setting living room temperature."
- Kitchen Lights: "Turning on bright task lighting."
- Kitchen Thermostat: "Setting kitchen temperature."

Write a program that takes these two character inputs (room and action) and uses a nested switch statement to print the appropriate command for the smart home system.

Student Name: _____Wajiha Nazir_____Roll No: _____25K-3021_____

Section: BSE-1B

```
#include<stdio.h>
int main()
{
    char room, action;
    printf("L=Living Room\nK=Kitchen\nEnter Room:\t");
    scanf(" %c",&room);
    printf("L=Lights\nT=Thermostate\nEnter Action:\t");
    scanf(" %c",&action);
    switch(room)
    {
        case 'L':
        {
            switch(action)
            {
                case 'L':
                    printf("Adjusting ambient lighting.");
                    break;
                case 'T':
                    printf("Setting living room temperature.");
                    break;
            }
            break;
        }
        case 'K':
        {
            switch(action)
            {
                case 'L':
                    printf("Turning on bright task lighting.");
                    break;
                case 'T':
                    printf("Setting kitchen temperature.");
                    break;
            }
            break;
        }
    }
    return 0;
}
```

```
L=Living Room
K=Kitchen
Enter Room:    L
L=Lights
T=Thermostate
Enter Action:  L
Adjusting ambient lighting.
-----
Process exited after 7.681 seconds with return value 0
Press any key to continue . . .
```

Student Name: Wajiha Nazir Roll No: 25K-3021

Section: BSE-1B

```
L=Living Room
K=Kitchen
Enter Room:    L
L=Lights
T=Thermostate
Enter Action:  T
Setting living room temperature.
-----
Process exited after 6.157 seconds with return value 0
Press any key to continue . . .
```

```
L=Living Room
K=Kitchen
Enter Room:    K
L=Lights
T=Thermostate
Enter Action:  L
Turning on bright task lighting.
-----
Process exited after 6.477 seconds with return value 0
Press any key to continue . . .
```

```
L=Living Room
K=Kitchen
Enter Room:    K
L=Lights
T=Thermostate
Enter Action:  T
Setting kitchen temperature.
-----
Process exited after 5.004 seconds with return value 0
Press any key to continue . . .
```

Question:7

A program checks user permissions stored in a single integer variable permissions. The value is a combination of flags: 1 (binary 001) for READ, 2 (binary 010) for WRITE, and 4 (binary 100) for EXECUTE.

The access rules are:

- If a user has the EXECUTE permission, print "Access granted: full control".
- If a user does NOT have EXECUTE but has both READ and WRITE, print "Access granted: read and write".
- If a user has only READ, print "Access granted: read-only".
- Otherwise, print "Access denied".

Write a nested if-else statement that uses **bitwise operators** to check these bits in permissions and prints the correct message.

```
#include <stdio.h>

int main() {
    int permissions;
    printf("Enter permissions value (1=READ, 2=WRITE, 4=EXECUTE, combinations allowed): ");
    scanf("%d", &permissions);
    if (permissions & 4) {
        printf("Access granted: full control\n");
    }
    else {
        if ((permissions & 1) && (permissions & 2)) {
            printf("Access granted: read and write\n");
        }
        else {
            if (permissions & 1) {
                printf("Access granted: read-only\n");
            }
            else {
                printf("Access denied\n");
            }
        }
    }
    return 0;
}
```

```
Enter permissions value (1=READ, 2=WRITE, 4=EXECUTE, combinations allowed): 4
Access granted: full control

-----
Process exited after 2.083 seconds with return value 0
Press any key to continue . . .
```

```
Enter permissions value (1=READ, 2=WRITE, 4=EXECUTE, combinations allowed): 3
Access granted: read and write

-----
Process exited after 1.419 seconds with return value 0
Press any key to continue . . .
```

```
Enter permissions value (1=READ, 2=WRITE, 4=EXECUTE, combinations allowed): 1
Access granted: read-only

-----
Process exited after 2.757 seconds with return value 0
Press any key to continue . . .
```

```
Enter permissions value (1=READ, 2=WRITE, 4=EXECUTE, combinations allowed): 2
Access denied

-----
Process exited after 2.142 seconds with return value 0
Press any key to continue . . .
```